

Professional Certificate in AI Engineering and MLOps: Using Cloud Resources Responsibly (AWS Guidelines)

Build smart. Manage costs. Stay in control.

As part of the Professional Certificate in AI Engineering & MLOps at **BITS Pilani Digital**, you will build and deploy real systems using **Amazon Web Services (AWS)**.

This is a key part of your learning experience—working in a real cloud environment, just like industry practitioners.

Since your AWS usage is **learner-funded-and-managed**, it's important to follow a few simple practices to **avoid unnecessary costs while still learning effectively**.

Note: Sign up to the AWS Account using your BITS Digital Email ID

1. Start with a Cost-Conscious Mindset

Cloud resources are powerful—but they are also **metered**.

- You are billed for **what you use, while it is running**
- Many services continue billing even when **idle but not stopped**
- Small oversights (like leaving instances running) can lead to **unexpected charges**

👉 Treat cloud usage like a **utility bill**—use what you need, and turn it off when you don't.

2. Always Stop or Terminate Resources

This is the **#1 rule for cost control**.

After every lab or project session:

- **Stop compute instances** (e.g., EC2, notebooks)
- **Terminate unused resources** when no longer needed
- Shut down:
 - Virtual machines
 - GPU instances
 - Notebook environments

⚠ Important:

“Closing your browser” does **NOT** stop billing.

3. Prefer Smaller, Cost-Effective Configurations

Start small. Scale only when needed.

- Use **low-cost instance types** wherever possible
- Avoid high-end GPU/compute unless explicitly required
- Test your code locally before deploying to cloud

👉 Most learning tasks do **not require expensive infrastructure**

4. Monitor Your Usage Regularly

Stay aware of your spend in real time.

- Use AWS **Billing Dashboard**
- Enable:
 - **Budget alerts**
 - **Usage notifications**

Set a simple rule for yourself:

“If I don’t understand a spike in cost, I investigate immediately.”

5. Clean Up After Every Experiment

Temporary resources can quietly accumulate costs.

Make it a habit to:

- Delete unused:
 - Storage volumes
 - Snapshots
 - Old deployments
- Remove test environments once validated

👉 If you don’t need it tomorrow, **delete it today**

6. Be Mindful of Storage and Data Transfer

Costs don't just come from compute.

- Large datasets stored long-term can add up
- Data transfer (especially across regions) may incur charges

Best practices:

- Store only what you actively need
 - Avoid duplicating datasets unnecessarily
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7. Follow Program-Recommended Setups

Wherever possible:

- Use **guided configurations** provided in the program
- Follow **instructor-recommended architectures**

These are designed to:

- Work effectively
 - Stay within **reasonable cost bounds**
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8. Learn Efficiently, Not Expensively

The goal is not to spend more—it's to **learn better**.

- Plan your sessions before starting resources
 - Avoid “trial-and-error” on expensive infrastructure
 - Use cloud time for **execution, not exploration**
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9. When in Doubt, Pause and Check

If you're unsure about a resource:

- Pause before launching
- Check estimated pricing
- Ask for guidance

A 2-minute check can save **hours of unnecessary cost**

Choosing the Right AWS Region (Cost Tip)

When working on **Amazon Web Services (AWS)**, the **region you select can impact your costs**.

General Guideline

- AWS pricing varies by region due to infrastructure and demand differences
- In most cases, **US regions are among the cheapest**

Recommended Low-Cost Regions

Learners can typically expect lower pricing in:

- **US East (N. Virginia)** (*us-east-1*) → often the **cheapest and most widely supported**
 - **US West (Oregon)** (*us-west-2*) → also relatively low-cost
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Regions to Be More Careful With

- Regions closer to India (e.g., **Mumbai, Singapore**) are often:
 - Slightly **more expensive**
 - But may offer **lower latency**
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What Should You Choose?

- For most labs and projects:

Use US East (N. Virginia) for lowest cost

- Only choose other regions if:
 - Specifically required in instructions
 - You need region-specific services
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Important Note

- Pricing differences are usually **not massive**, but over time they add up
 - Always ensure the services you need are available in the chosen region
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Rule of thumb:

If cost is your priority → go with **US East (N. Virginia)**

If latency or instruction requires → use the specified region

Estimated Monthly Cloud Cost (AWS)

Plan ahead. Spend wisely.

As you work on labs and projects using **Amazon Web Services (AWS)**, your monthly costs will depend on how actively you use cloud resources and how efficiently you manage them.

Typical Monthly Spend Range

For most learners, the expected cost falls within:

₹1,000 – ₹3,000 per month (~\$12 – \$35 USD)

What Drives Your Costs

Your actual spend may vary based on:

- **Compute usage**
 - Running virtual machines (especially GPU instances)
 - **Duration of usage**
 - How long resources remain active
 - **Storage**
 - Datasets, model artifacts, backups
 - **Experimentation patterns**
 - Frequency of trials and deployments
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When Costs May Be Lower (~₹1,000)

- You:
 - Use **small instance types**
 - Shut down resources promptly
 - Follow guided labs closely
 - Minimize idle time

👉 Typical of **disciplined, planned usage**

When Costs May Be Higher (~₹2,000–₹3,000)

- You:
 - Run **multiple experiments**
 - Use **larger compute instances occasionally**
 - Spend more time on the capstone project

👉 Typical of **active builders pushing deeper**

When Costs Can Spike (Avoid This)

Costs may exceed ₹3,000 if:

- Instances (especially GPU) are left **running unintentionally**
- Unused resources are **not terminated**
- Large storage or multiple environments are left active

⚠️ These are usually **avoidable mistakes**, not necessary spend.

Recommended Budgeting Practice

To stay in control:

- Set a **monthly budget cap of ₹1,500**
 - Enable AWS **billing alerts at ₹1,000 and ₹1,250**
 - Review usage **weekly**
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AWS Promotional Credits (\$200 for New Users)

New users on **Amazon Web Services (AWS)** may receive **promotional credits (up to \$200)** as part of their account onboarding.

What This Means

- These credits can be used to **offset usage costs** across many AWS services
 - Typically valid for a **limited duration (e.g., ~6 months)** or until the credits are exhausted—whichever comes first
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How to Use This Effectively

- Treat the credits as a **learning buffer**, not a spending target
 - Use them for:
 - Initial labs
 - Early experimentation
 - Exploring services safely
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Important Considerations

- Credits:
 - **Do not cover all services** in all cases
 - **Expire after the validity period**, even if unused
 - Once credits are exhausted:
 - **Regular billing begins automatically**
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Common Misunderstanding

“I have \$200 credits, so I won’t be charged.”

This is **not guaranteed**:

- High-cost resources (e.g., GPUs, long-running instances) can **consume credits quickly**
 - Improper resource management can still lead to **out-of-pocket expenses**
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Recommended Approach

- Use credits **strategically in the first few months**
 - Continue to follow all **cost-control best practices**
 - Track:
 - Remaining credits
 - Monthly usage trends
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Rule of Thumb

Think of AWS credits as **training wheels—not a free ride**

They help you get started safely, but **cost awareness is still essential**.

Final Thought

The goal is not to minimize spend at all costs—it's to:

Maximize learning while staying efficient

In real-world AI engineering, managing infrastructure cost is just as important as building the system itself.

A Final Note

Working with AWS is an opportunity to:

- Build **real-world, deployable systems**
- Develop **cost-awareness as an engineer**
- Learn how to balance **performance and efficiency**

Managing cloud costs effectively is not just a precaution—it's a **core engineering skill**.